
Revision 1 Documentation:

Requirements, Hazards Analysis, Design, Validation, and Verification

Robotic Intervention for Pollution and Plastic Litter Extraction

(RIPPLE)

MECHTRON 4TB6: Mechatronics Capstone Design Process

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Group 14

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Revision History

Table 1: Revision 1 Changes Table

Date	Section	Details of Change	Related TA/ Supervisor Feedback
March 12 th , 2026	All Sections	Diagrams changed from .jpg/.png to .svg files	TA – “Change diagrams to SVG files so they don’t get funky when zooming in.”
March 12 th , 2026	All Sections	Added proximity sensor information	N/A
March 13 th , 2026	Hazards & Requirements	Updated requirements and hazards with determined testing values	TA – “Ensure consistency” added to Undesired Events Handling
March 13 th , 2026	Verification & Validation	Retested and documented results of previously failed motor control tests	TA – “Change design so it can pass failed tests”
March 14 th , 2026	System Design	Updated motor control design to include motor cooling method	TA – “Change design so it can pass failed tests in V&V”
March 14 th , 2026	System Design	Updated UI code to integrate GPS coordinates and move marker on map	TA – “Change design so it can pass failed tests in V&V”
March 14 th , 2026	System Design	Updated GPS design information	N/A

Purpose of Document

The purpose of this document is to evaluate the hazards that can arise from a remote-controlled surface-water trash collecting system and propose safety, behavioural, and technical requirements to achieve desired system outcomes and mitigate the modes of failure identified in the hazard analysis. The document will further detail the design of the system to satisfy defined behavioural and safety requirements. Additionally, the document will detail the testing plan and results to determine if the designed system satisfies defined behavioural and safety requirements.

Scope of Project

RIPPLE is a robotic system that is designed to collect small floating trash on the surface of local ponds. It will be remotely controlled by the user and communicate its location back to a UI. A camera stream will also be displayed on the UI to allow the user to identify obstacles in the system's path.

The system is intended to be used by a trained adult familiar with the controls in a pond or other small body of water that has accumulated trash on its surface in calm, non-precipitating weather. The project is intended to collect the floating trash as directed by its operator.

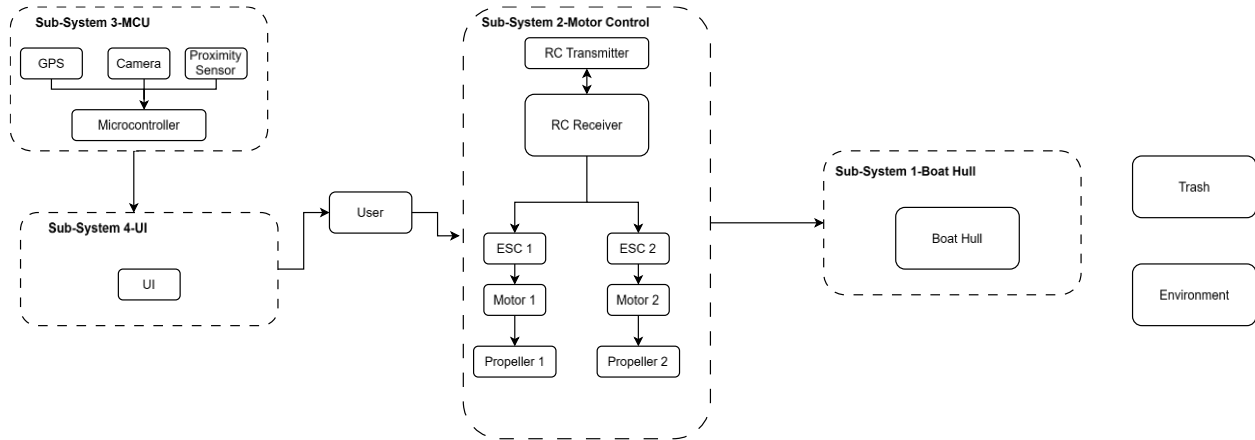
Definitions

Table 2: Definitions

Word/Acronym	Definition
Boat Hull	This is Sub-System 1. This includes the 3D printed boat and net mechanism.
Motor Control	This is Sub-System 2. This includes the motors, ESC, propellers, and RC.
MCU	This is Sub-System 3. This includes the microcontroller, GPS module, camera module, and proximity sensor.
UI	This is Sub-System 4. This is the User Interface.
System	This is the entire system including all Sub-Systems.
TCM	Trash Collection Mechanism. This includes the boat hull and all parts and components attached to it. (All components moving with the boat hull)
RC	Remote/Radio Controller
RID	Requirement Identifier Number
SSR	Safety & Security Requirement
Trash	Floating trash on the surface of the pond that is within the size constraints of the TCM.
ESC	Electronic Speed Controller
HID	Hazard Identifier Number
UM	User's Manual
RPM	Revolutions Per Minute

The System

Figure 1: System Diagram Showing Boundaries



“The System” as described in this document includes four major sub-systems that interact with the user. Sub-system one includes the Boat Hull, which is the 3D printed parts of the system’s body. Sub-system two is the Motor Control and its primary goal is to transform user inputs into desired movement, achieved through an RC connected to the boat’s propellers. The third sub-system includes the GPS, Camera modules and the Proximity Sensor controlled by a microcontroller whose main functions are to provide feedback on the system’s state. This is connected to the fourth sub-system which is the UI, whose purpose is to serve as a visual guide for the user to move the boat. Other important elements of the project that fall outside the controllable boundaries of the system include the trash the system wishes to collect, and the system’s environment.

Monitored & Controlled Variables

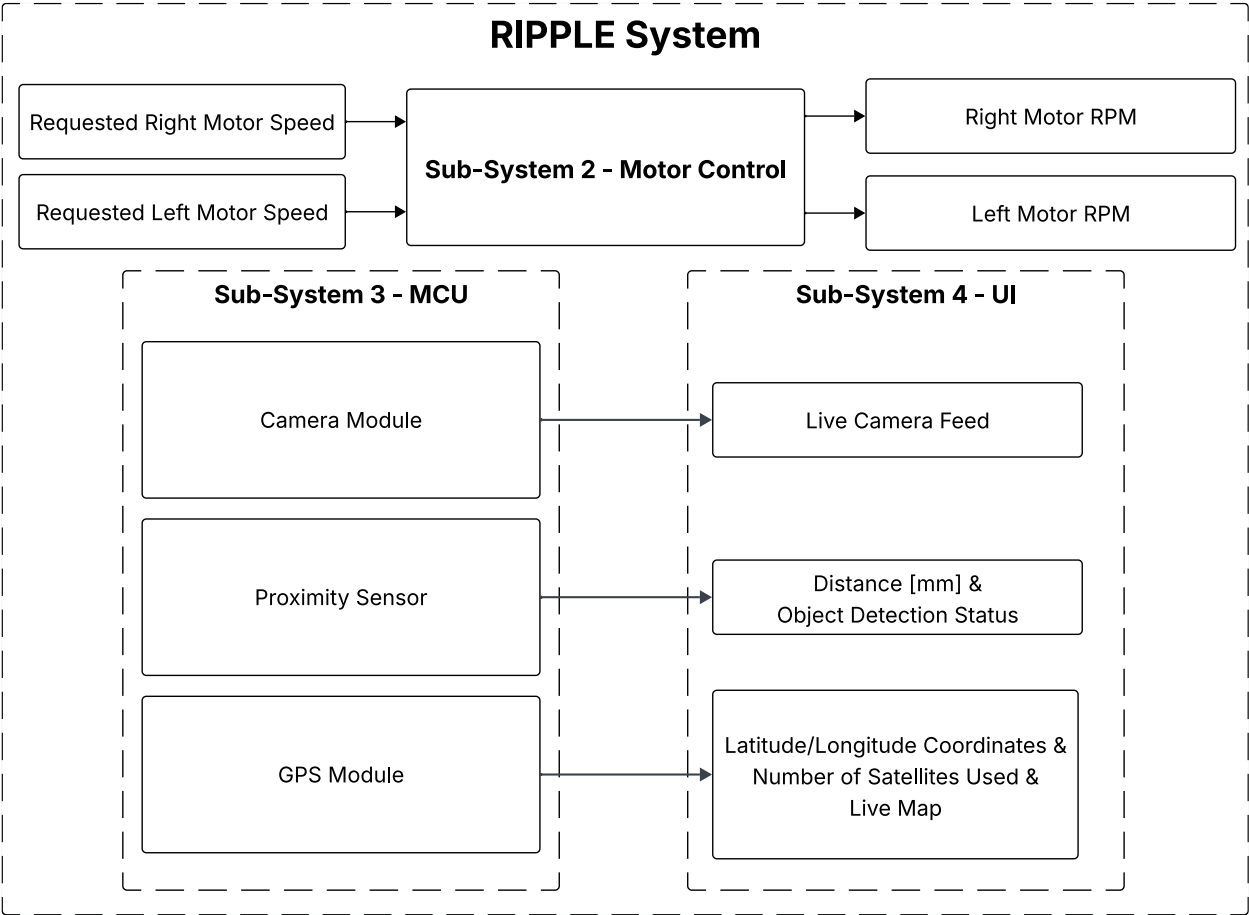
Table 3: Monitored Variables

Monitored Variables (System Inputs)	Description
Vertical Joystick Angle	This corresponds to clockwise and counterclockwise motor rotation resulting in forward and backwards TCM motion respectively, activated by moving the controller joystick forwards and backwards on the RC.
Horizontal Joystick Angle	This corresponds to right and left motor isolated activation resulting in TCM rotation, triggered by moving the controller joystick left and right on the RC.

Table 4: Controlled Variables

Controlled Variables (System Outputs)	Description
Right Motor RPM	This is the actual right motor speed that will turn the right propeller.
Left Motor RPM	This is the actual left motor speed that will turn the left propeller.
GPS Coordinates	These are the x and y coordinates of the system, that will be updated in real-time and be displayed on the UI.
Live Camera Feed	This is the live camera feed that will be updated in real-time and be displayed on the UI.
Proximity Sensor	Indication of trash flowing towards the TCM.

Figure 2: Context Diagram



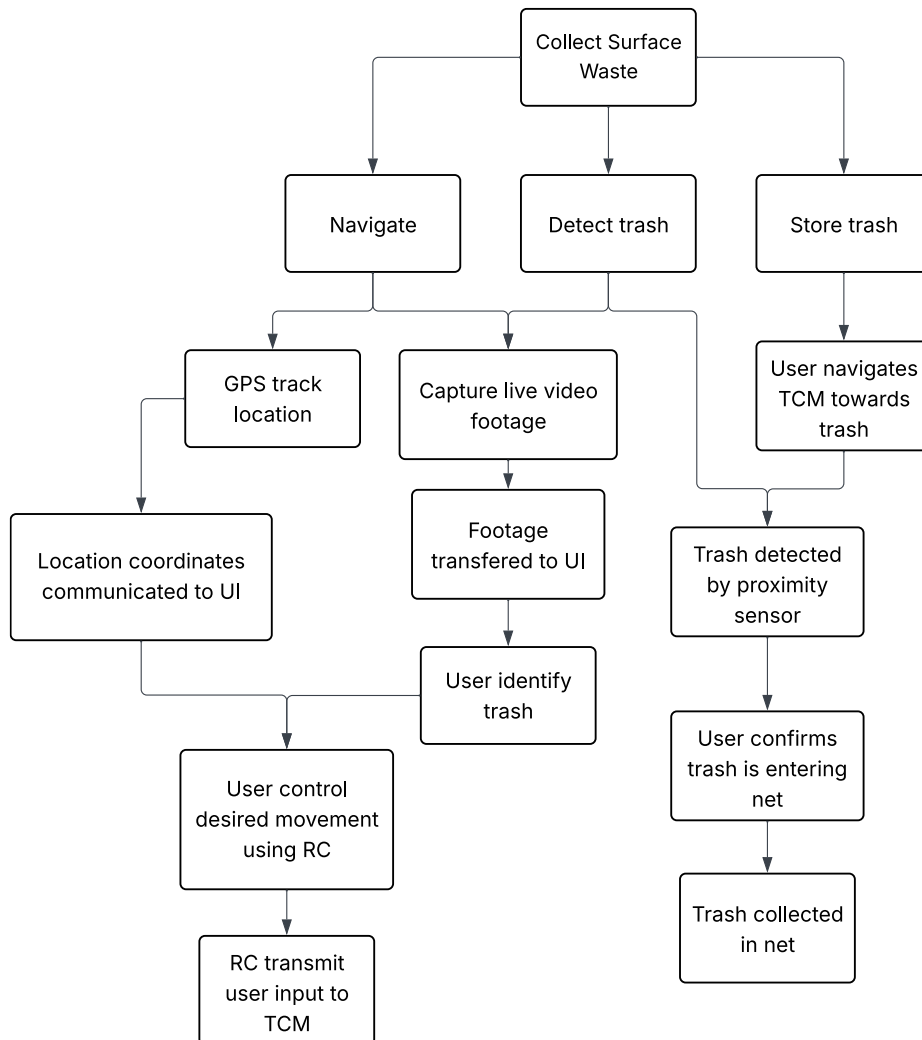
Constants with Nominal Values & Ranges

Table 5: Constants with Nominal Values & Ranges

Constants	Nominal Values & Ranges	Units
Maximum Distance from User	The minimum of either the RC range or the Wi-Fi range of the microcontroller.	meters [m]
Maximum Speed of Motors	5 m/s; based on motor specifications and testing.	meters per second [m/s]
Maximum Weight on System	2240g; based on testing.	grams [g]
Maximum Transmission Delay (RC)	Maximum time delay between transmission of RC controller data and receipt of said data by the receiver (on boat).	microseconds [us]
Maximum Transmission Delay (MCU)	Maximum time delay between collection of data from GPS and camera to receipt of data by the host PC (user).	microseconds [us]
GPS and Proximity Sensor Poll Period	1s; time between each subsequent request and/or receipt of GPS and proximity sensor data.	seconds [s]

Functional Decomposition

Figure 3: Functional Decomposition Diagram



System Hazards

Critical Assumptions

1. The system is being operated by a person with the required competence who is familiar with the UM.
2. The system is being used in its intended environment and weather conditions without any strong wind gusts.
3. The system is being used in the appropriate body of water with appropriate water depth and minimal to no waves, such as a pond.
4. There are no motorized vehicles or human beings in the body of water.
5. The system's battery is fully charged, camera lens is clean, and there is no damage prior to deployment onto the water.
6. The TCM is configured correctly and empty prior to deployment onto the water.

Hazards Analysis – Fault Tree Analysis

Figure 4: Top Level Hazards

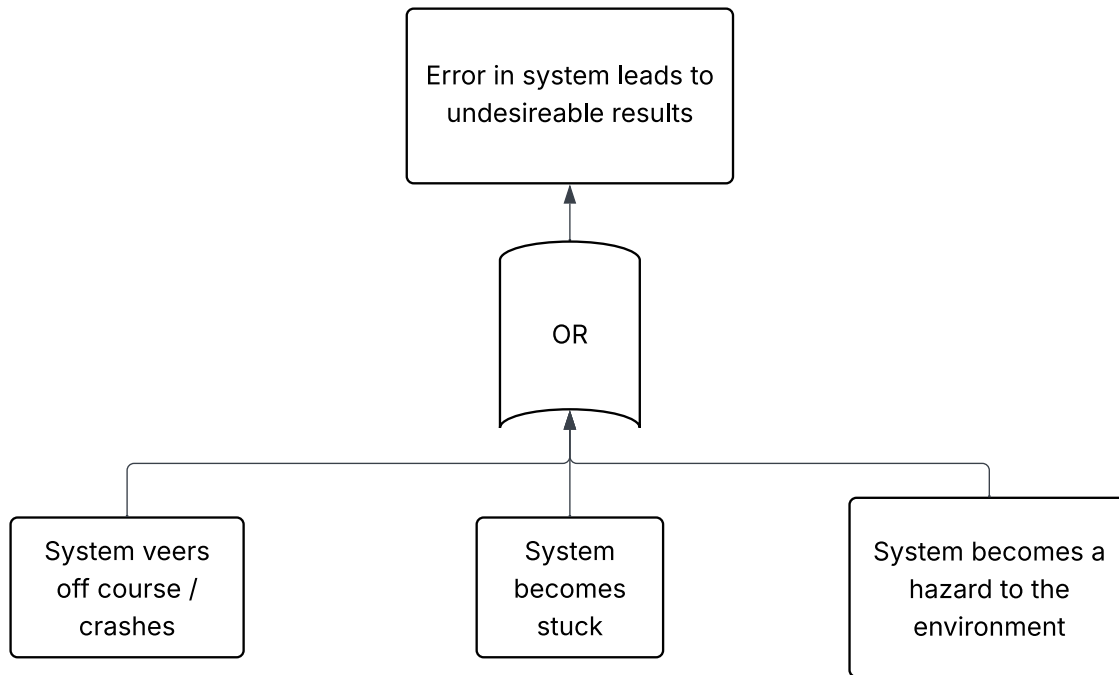


Figure 5: Second Level Hazards 1

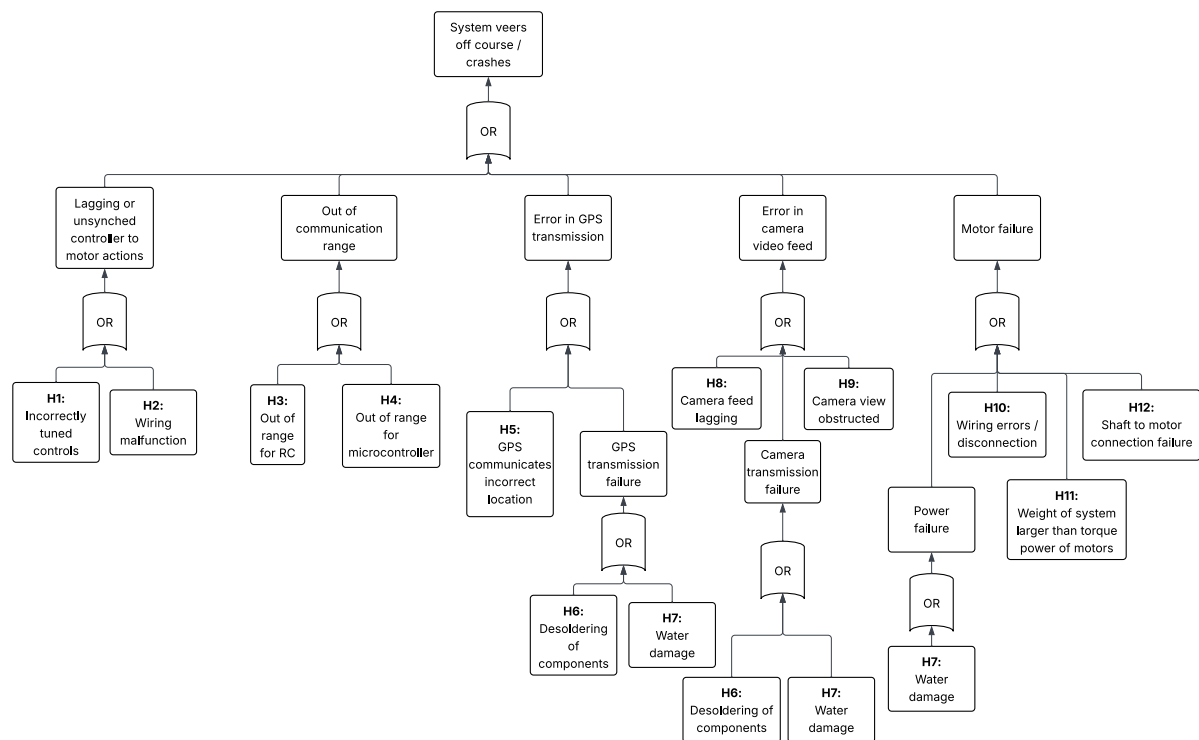


Figure 6: Second Level Hazards 2

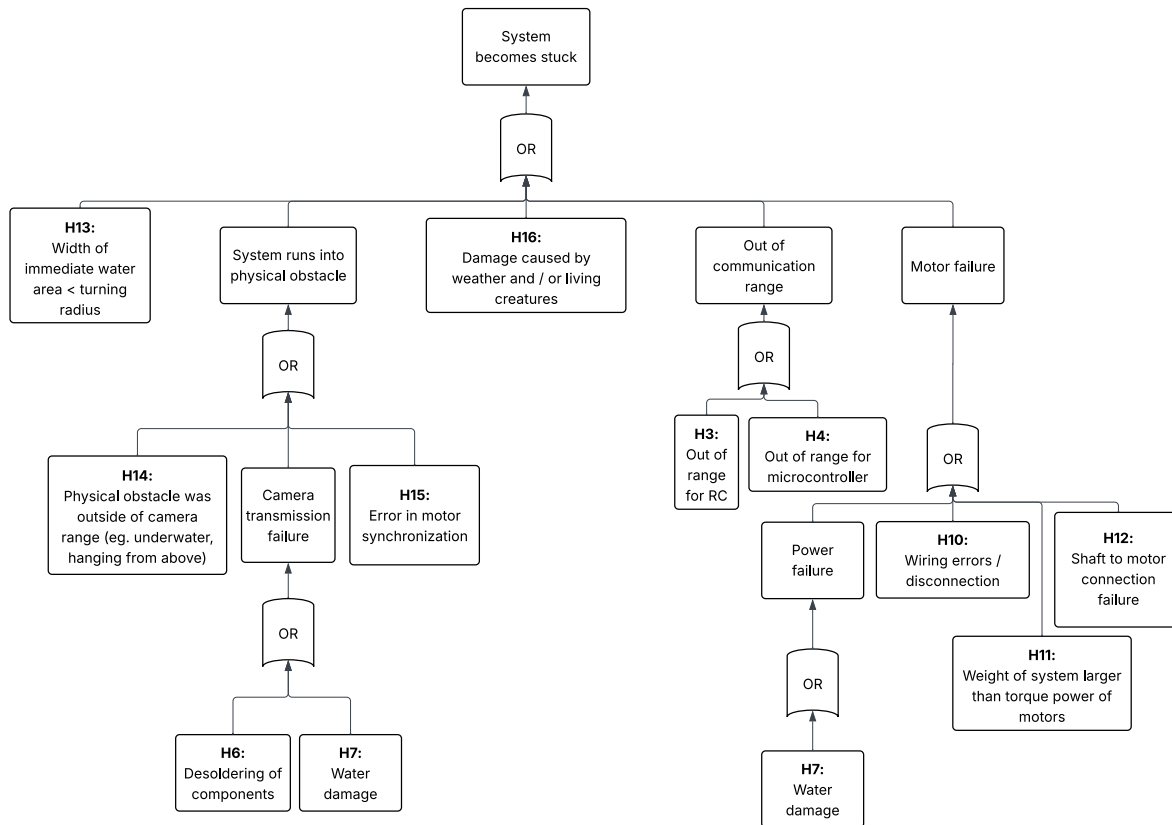
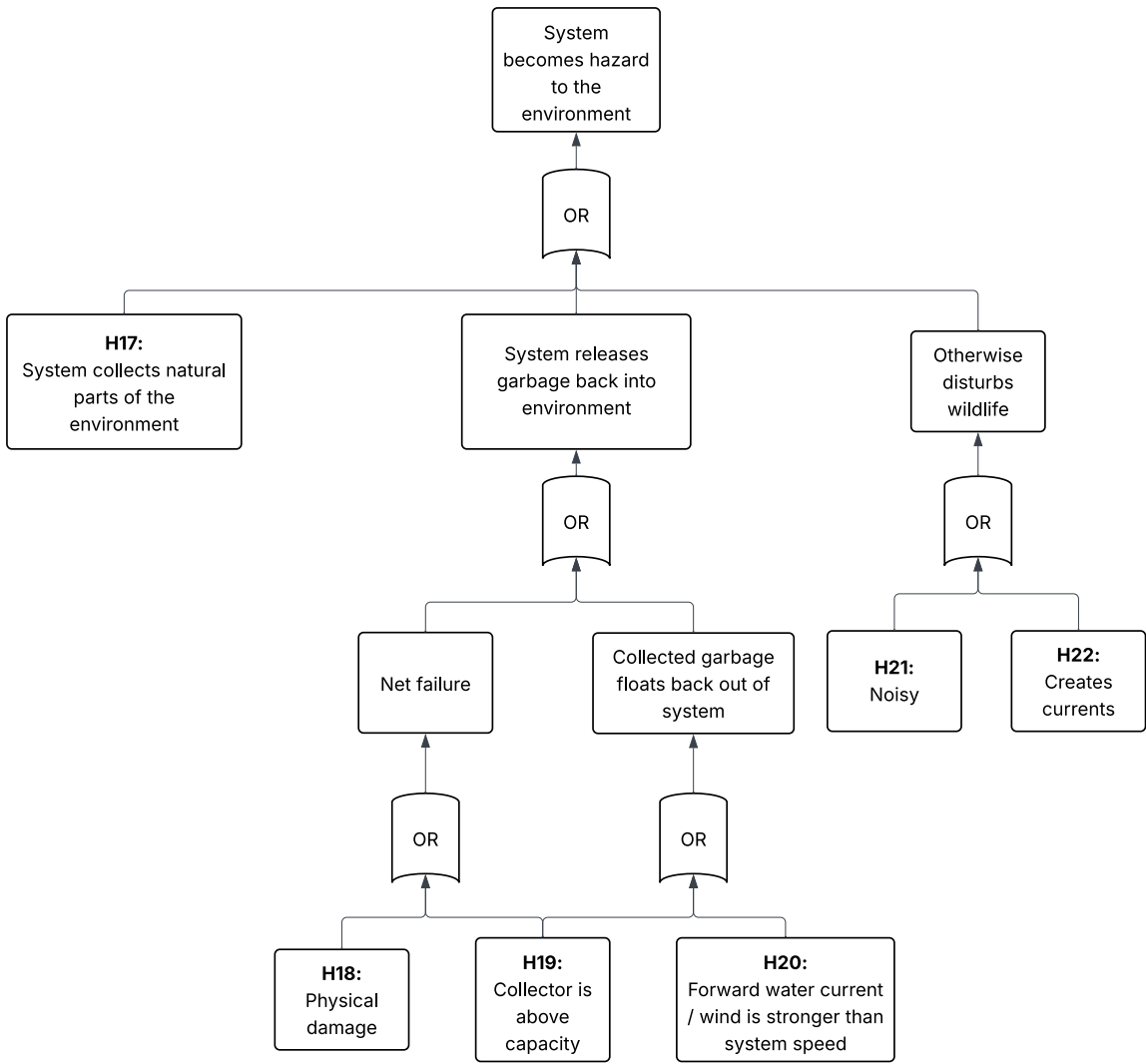


Figure 7: Second Level Hazards 3



Risk & Mitigation Strategies

Table 6: Risk & Mitigation Strategies

HID	Risk Type	Description of Risk	Mitigation Strategy
H1	Incorrectly tuned controls	Incorrect tuning of the control system can lead to unpredictable behavior of the system such as failure to respond to control input as needed.	Rigorous tuning of controls during development process. Consider all edge cases.
H2 H15	Motor synchronization error	The motor being unsynchronized with the RC controller may cause loss of directional control.	Test motors beforehand to find the minimum distance at which synchronization with the RC is lost. This strategy is extracted from the RC's user manual.
H3	Out of RC range	If the system travels out of the bounds of the RC range this will lead to loss of signal thus the user will no longer be able to direct the system.	Motor synchronization errors are expected to only occur when out of range. So, strategy used to address motor synchronization also addresses this hazard.
H4	Out of MCU range	If the MCU is too far from the hosting server communication may be lost, causing control delays and the system being disconnected from the server.	Testing with MCU (client) and host PC (host), and consult MCU specifications, to find the minimum distance at which synchronization is lost. Indicate this distance as the absolute maximum range of the TCM in user manual.
H5	GPS communicates incorrect info	The GPS module could output delayed position data, which can lead to navigation errors.	Test and check all relevant and possible edge cases to guide development of GPS handling algorithm.
H6	Desoldering of components	Mechanical stress, heat, or poor assembly can cause electrical components to desolder, leading to a system failure.	Ensure the components are soldered properly prior to TCM assembly.
H7	Water damage to electrical components	Exposure to water may damage sensors, wiring, or the MCU, which can lead to system failure.	In user manual, include a warning that electrical components could be damaged. Use waterproof electrical components which will be evaluated to ensure waterproofness before placing electronics in the boat hull compartment.
H8	Camera feed lagging	Delayed camera feed or signal loss can affect the operator's ability to see and control the robot accurately.	Test extensively beforehand, and consult camera and MCU

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HID	Risk Type	Description of Risk	Mitigation Strategy
			specifications, to find maximum delay in camera transmission.
H9	Camera view obstructed	External objects may block the camera's view, reducing situational awareness and increasing collision risk.	In user manual, include a warning of this possibility and direct users to be attentive of potential obstructions during operation. The user may see in the camera view area obstructed and should bring the boat back home.
H7 H10	Motor power failure	The motors may stop functioning because of electrical faults, mechanical failure, or overload which will lead to the TCM stalling or immobility.	If GPS detects no change in position of TCM for 30 seconds during operation, system will give user a warning that motors are not functional.
H10	Disconnected wires	Improper assembly may cause wires to disconnect, preventing power delivery or signal transmissions to critical components.	Ensure wires are well-secured throughout the system.
H11	TCM mass > motor torque power	If the total weight of the TCM exceeds what the motors can handle, the robot may fail to maneuver properly.	Verify during testing that the total TCM mass does not exceed the motors' rated torque output. Adjust design by using lighter materials if necessary to stay within safety limits.
H12	Shaft to motor connection failure	The connection between the motor and propeller shaft may loosen or break, resulting in loss of propulsion on one or both sides.	Secure motor well to propeller shaft using strong adhesives and warn user manual of risk of detachment.
H13	Water area width < turning radius	The TCM may be operated in an area too narrow for proper turning, increasing the likelihood of getting stuck.	Include risk in user manual and warn users to be cautious of this possibility.
H14	Physical obstacle outside camera range	Objects not visible in the camera's field of view may pose a collision risk.	Include risk in user manual and instruct users to operate the TCM in an area which does not have many low-overhanging branches. Proximity sensor to be used to prevent this risk.
H17	TCM collecting natural environment	TCM may collect natural material that are a part of the ecosystem reducing capacity and affecting system performance.	Include risk in user manual and warn users to be cautious of this possibility. Camera to be used for object detection to mitigate this risk.

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HID	Risk Type	Description of Risk	Mitigation Strategy
H16 H18	Physical damage to net	The collection net may tear from sharp debris or excessive load, leading to loss of collected waste.	Include risk in user manual and warn users to be cautious of this possibility.
H19	Net above capacity	Overloading the net may reduce maneuverability, strain the motors, or cause the net to tear or detach.	Include net's capacity in user manual and warn users to use their best judgement during operation to ensure the net is regularly emptied to not exceed capacity.
H20	Forward water current/wind stronger than TCM speed	Strong currents or winds may push the TCM off course, reducing control.	Include warning in user manual not to operate in extreme weather conditions.
H21	Noisy	The motors or mechanical components may produce excessive noise, potentially disturbing wildlife or violating noise expectations.	Use sound-insulating/thicker materials around noisy areas of the TCM internals to muffle noise.
H22	TCM creating currents	The TCM's propulsion may generate disruptive water currents that affect its stability or environmental conditions.	Limit maximum operating speed of TCM and include in user manual.

System Requirements

Likelihood of Changing: How probable the requirement will be revised as development progresses. Below are the definitions of the low, medium, high likelihood of changing and will be added to all requirements tables below.

Table 7: Definition of Requirement's Likelihood of Change

Low	The requirement is very stable, meaning it is highly unlikely to change.
Medium	The requirement is somewhat stable, meaning minor adjustments may occur.
High	The requirement is unstable meaning it is most likely going to change.

Safety & Security Requirements Derived from Hazards

The following requirements identify key failure modes such as loss of control, navigation errors, communication failures, and external environmental hazards. They are derived from the root causes of hazards and are classified as Safety & Security Requirements (SSR).

Table 8: Safety & Security Requirements

RID	HID	Description	Likelihood of Changing
SSR1	H7	Waterproof system: The electrical components in the TCM shall have waterproof housings to prevent damage from splashes.	Low
SSR2	H7 H10 H11 H12	Emergency stop: If TCM loses connection with RC for more than 5 Seconds, power supply shall be cut off.	Low
SSR3	H14	Alert for Environmental Interference: TCM should alert user when motor is moving but GPS coordinates are not updating. <i>Rationale: Allows user to check for environmental interference (animals / branches / rocks / stuck / etc.)</i>	Medium
SSR4	H3 H4	Communication Range: TCM should alert user through UI when TCM is 3 meters away from out-of-range boundaries.	Medium
SSR5		User Manual: User manual shall be created to prevent operational hazards.	Low
SSR5.1	H20	<i>User must operate system in ideal weather and water conditions.</i>	Low
SSR5.2	H9	<i>User must ensure that camera is clean / view is not obstructed before operation.</i>	Low
SSR5.3	H17	<i>User must monitor the camera footage rigorously during operation.</i>	Low
SSR5.4	H16 H18	<i>User must check system upon recollection for physical damage.</i>	Low
SSR5.5	H19	<i>User must be wary of the amount of trash collected by the system to mitigate overflow.</i>	Low
SSR5.6	H13	<i>User must be aware of max turning radius.</i>	Low
SSR6	H22	Speed Limit: Maximum operating speed of TCM shall not exceed 5 m/s.	High
SSR7	H21	Sound Insulation: System should be sound insulated so as not to disturb the natural environment.	Medium
SSR8	H1 H2 H5 H6 H8 H10 H15	Component Validation: All electrical components of system shall be tested prior to system assembly to validate specifications.	Low

Requirements Specification

The following requirements are the high-level behavioral requirements of the system.

Table 9: Behavioral Requirements

RID	Description	Likelihood of Changing
R1	Floatation: The TCM shall be floating during the entirety of the operation cycle.	Low
R2	Trash Collection: The TCM shall securely collect trash floating on the surface of small bodies of water.	Low
R3	Optimal Operation: The TCM shall operate reliably on the surface of water carrying all associated components and any collected trash up to 1000 cm ³ .	Low
R4	System Movement: The TCM's movement should be remotely controlled by the user.	Medium
R5	Turning Radius: The system should allow forward movement and changes of direction with a 30 cm turning radius.	Medium
R6	Location Detection: The TCM component should be remotely located by the user using GPS coordinates.	Medium
R7	Visualization: The system shall provide the user with live video footage of the front view of the TCM.	Low

Performance Timing Requirements

The following requirements define the performance expectations of our surface-water trash collector, specifying how quickly the system must detect, collect, and dispose of floating debris in West Pond.

Table 10: Performance Timing Requirements

RID	Description	Likelihood of Changing
R8	Detection Response Time: Using camera visualization, the camera visual should be able to detect a standard floating object approaching the TCM within 5 seconds.	Low
R9	Collection Cycle Time Per Item: The TCM should complete a full collection time under normal conditions (camera detect, align, capture, store) for a properly sized item in less than 20 seconds.	Medium
R10	Battery Efficiency Cycle: The system shall operate continuously for at least 1 hour before it needs to be recharged.	Low
R11	Alignment Time: The TCM should properly align the net to intercept an object within 10 sec from 2 meters away from the net.	Medium
R12	Direction Change Time: The TCM should be able to change direction by 90-degree angle in 3 seconds.	Low
R13	Thrust to Drag Response Time: The propulsion mechanism shall generate sufficient thrust to overcome drag and provide forward motion within 3 seconds of motor activation from rest. Rationale: This ensures that the system can overcome water and collect trash resistance quickly, preventing stall at low speeds.	Low
R14	Motor Response Time: Each of the motors should reach 100% of the user input speed within 3 seconds of receiving the control signal.	Low
R15	GPS Response Time: The delay between receiving GPS data from the GPS and the ESP32 main control loop processing shall be $\leq 1\text{sec}$	Low

Requirements for Hardware Environment

These requirements below define the hardware requirements for the surface-water trash collector, specifying the physical components and equipment needed for the system to function properly.

Table 11: Hardware Requirements

RID	Description	Likelihood of Changing
R16	Operation Temperature: The system should operate reliably between 3 - 28 degrees Celsius.	Medium
R17	UV-Stabilization: The TCM's plastic enclosures, plastic structure, and net must be UV-stabilized to limit degradation.	Low
R18	Operation Levels: The system's safe operation is guaranteed up to wind $\leq$ TBD , wave height $\leq$ TBD , current $\leq$ TBD m/s.	High
R19	Dynamic Load Operation: Net frame and attachments shall withstand the impact of dynamic loads (drag force) and hold up to 500 g continuous load in the TCM.	Low
R20	Maintenance Accessibility: The system's compartments (battery, electronics, motors, GPS tracker, camera) shall be accessible for inspection/maintenance.	Low
R21	Component Mounting: GPS Tracker and camera shall be mounted on a raised, unobstructed mount.	Low
R22	Camera Specifications: Camera viewport shall be optically clear. Camera must support required resolution for detection.	Low
R23	Trash Loss Prevention: The TCM should be in continuous forward motion to prevent the loss of trash collected.	Medium
R24	Connection Status: The system shall have sufficient wireless connection between the ESP32 and the controller.	Low

Requirements for Software Environment

The requirements below define the software requirements for the surface-water trash collector, specifying the microcontroller and the UI.

Table 12: Software Requirements

RID	Description	Likelihood of Changing
R25	Real-time Control: The remote controller shall enable real-time control of the TCM's movement, allowing independent operation of the two motors.	Low
R26	Sensor Integration: The software shall interface with the onboard GPS module and live camera to collect position and visual data.	Low
R27	Data Communication: The system shall transmit data from the microcontroller to the external UI.	Low
R28	UI Support: The UI should display real-time TCM coordinates and live video feed.	Medium
R29	Synchronization and Timing: The software shall ensure synchronized operation between remote controller data, motor control, and data transmission.	Low
R30	Modularity: The software shall be structured in modules (motor control, GPS, camera, UI) to simplify development, testing, and maintenance.	Low
R31	Extensibility: The software environment should allow integration of additional modules or functionalities such as mapping, analytics, or remote monitoring.	High

List of Requirements That Are Likely to Change

All requirements that have *High* or *Medium* Likelihood of Changing are likely to change. Most of the requirements are shown to have values that are undetermined which are subject to change as we continue to proceed with the project. These values will be determined during the testing phase and are subject to change. For example, the performance timing requirements are the most likely to change because they depend on environmental factors, such as water conditions, battery levels, and trash load. Variations in weight, drag, and available thrust during testing can affect system response, making these requirements more sensitive and subject to adjustment. However, the format of these associated requirements will most likely not change.

In addition, if it is decided that navigation control will be centralized within the UI, rather than controlled by the RC, it is likely that additional requirements related to motor failure detection will be added.

List of Requirements That Are Not Likely to Change

All requirements that have *Low* Likelihood of Changing are not likely to change. The requirements that are unlikely to change are those without a defined value range such as ensuring the camera is mounted without obstruction, which is an obvious necessity, and that the boat remains continuously afloat. These are requirements that will need to be implemented to ensure our system successfully collects trash.

System Design

Behaviour Overview

The integrated surface-water trash collecting system will securely collect trash floating on the surface of small bodies of water while operating reliably carrying all associated components and any collected trash. The TCM's movement will be remotely controlled by a user allowing forward, backward, and rotational direction changes. The system will inform users of the TCM's current location and provide live video footage of its front view to allow for optimal control decisions.

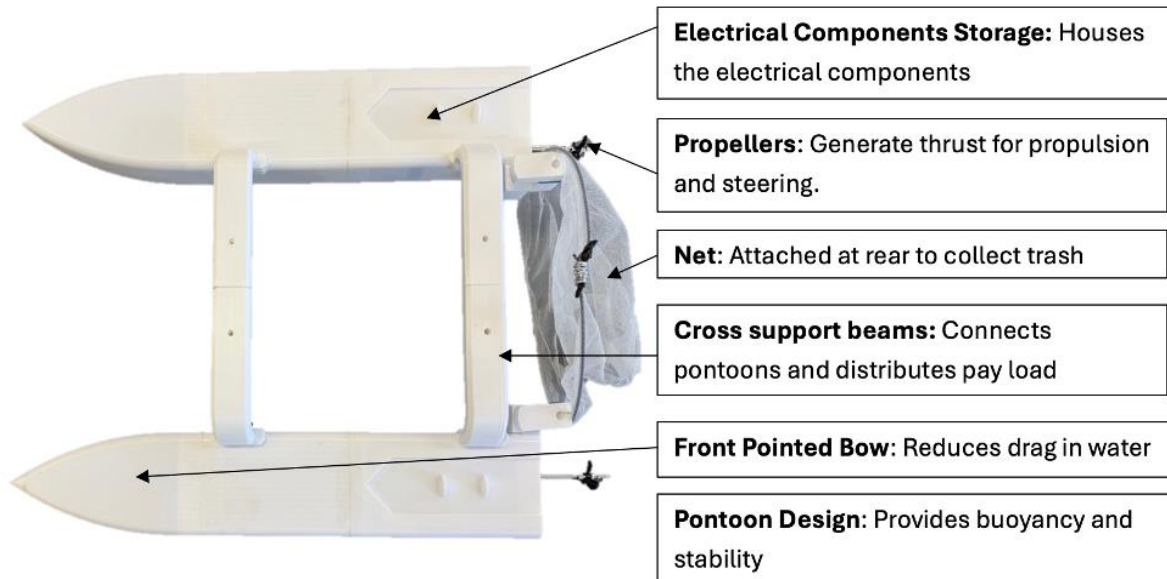
Sub-System 1: Boat Hull

Design

The boat hull design features twin-hull catamaran style boat, using two pontoons connected by two bridges. The pointed front end helps the boat cut into the water smoothly reducing drag and improving efficiency. The distance between the pontoons provides great stability, making the boat less likely to tip and allowing it to handle uneven movement in water. Inside these pontoons, there is ample enough storage for the motor and other electrical components including the propellers. The bridges allow the boat to have a flat desk area which will be used for storing components like the ESP-32, GPS, and battery, which is enclosed by a plate. There is a camera mount that is sitting on top of the plate and is angled appropriately to allow for better visualization of the front. Overall, the design focuses on stability, safety, smooth movement, and buoyancy.

Boat Hull Overview

Figure 8: Boat Hull Overview Diagram



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Table 13: Description of Boat Hull Components

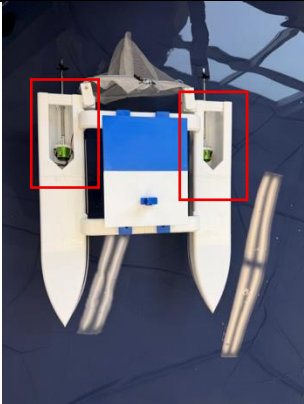
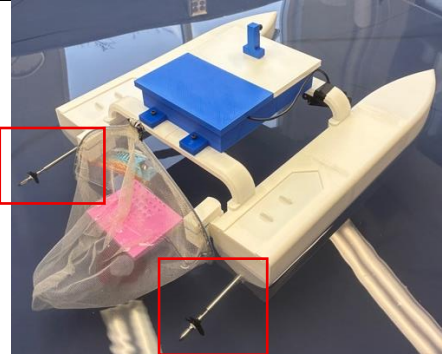
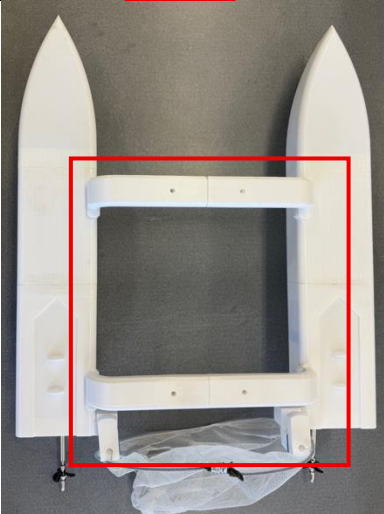
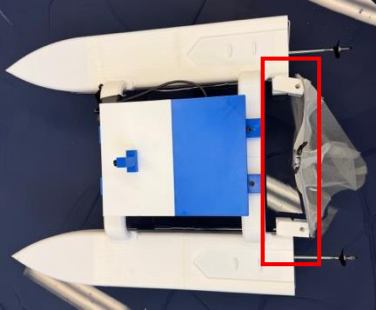
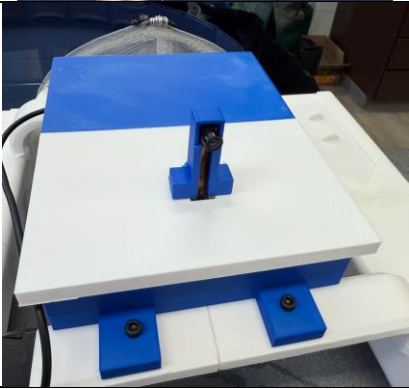
Image	Function
	Left/Right Body: Connected via a bridging mechanism, used to keep the TCM afloat. Fins that can be seen are used to stabilize the boat and direct water flow to help with steering
	Left/Right Compartment of Boat Hull: Used to store motor, battery, wires and other electrical components that is used to control the TCM
	Shaft: Attached to the motor used to create movement and control directions. The pivot joint decreases tension exerted on the motor
	Bridging Mechanism: Used to connect the left and right side of the boat hull. Designed to be above the water level to allow surface waste to pass through as the boat hull is moving across the water.

Image	Function
	Plate: Attached to the bridging mechanism, used as storage compartment for the micro controller, camara, and other electrical wires/components. Lids are placed on top to prevent water damage.
	Net Connection: Used for attaching the waste net to the back of the boat hull. As the TCM moves across the body of water, surface waste will be collected into the net.
	Camera Mount Camera housing angle camera 30 degrees downward for better visualization.

Inputs & Outputs

The inputs to the boat hull design include the payload, mass of the printed hull, safety factor, displaced water mass, and hull dimensions. These inputs are used to determine key performance outputs such as floatability, buoyancy, recovered trash capacity, boat hardware requirements (net frame, bridges, and side structures), propulsion performance, net collection rate, and operating speed based on the hull design.

Description of Behaviour

The boat hull is designed to support the necessary weight, so it can safely carry the electrical components required for propulsion. It must also generate enough thrust to overcome the drag created by the collected trash and continue moving forward. The boat hull is designed to provide sufficient buoyancy and stability to support all onboard systems. It keeps a low center of gravity for improved stability to allow for more predictable movement. The hull is designed to keep the center of gravity low, helping it remain stable and less affected by waves or shifting loads.

Testing

Testing of the floatability and waterproof performance is performed using a kiddie pool with a radius of 1.24 meters. The pool is located indoors, allowing testing to be carried out throughout the winter. The dimensions of the pool are significantly larger than those of the boat, which ensures sufficient testing space and helps with approximate real-world operating conditions.

Through testing, it was found that the boat hull can withstand a maximum load of 2.24 kg, which is more than sufficient to support all required components.

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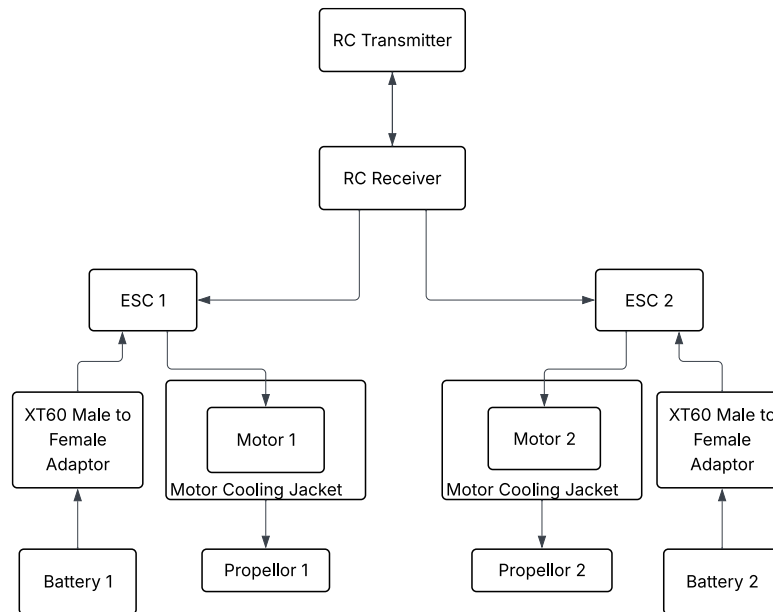
Figure 9: Pool Visual



Sub-System 2: Motor Control

Motor Control Structural Overview & System Behaviour

Figure 10: Motor Control System Structure



The system is comprised of an RC controller, and two sets of: ESCs, motors, propellers, XT60M to Deans T F adaptors, motor cooling jackets and battery packs. User control inputs are applied to the RC transmitter where they are communicated to ESC 1 (located on the right side of the boat) and ESC 2 (located on the left side of the boat) through RC receiver channels 2 and 5 respectively. These signals oversee controlling the rotary speed of the propellers at the stern of the boat hull through the RPM of two rotary motors. Power to the RC receiver, ESCs, and motors are provided by two battery packs and is distributed to all components through the ESC.

Inputs & Outputs

The inputs of Sub-System 2 are achieved via user input on an RC controller. Noted below are the relevant switches that the user will interact with, their input actions, the transmitter RC channels

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they are associated with, the receiver RC channels their output is communicated through, and what these control action achieve in the overall system.

Table 14: Functional Description of RC Controller Switches

Switch	Input	Input Channel	Output Channel(s)	Purpose
Right Joystick	Vertical Joystick Angle	N/A	Channel 2 (right motor) Channel 5 (left motor)	The vertical joystick angle controls the forward and backward movement of the TCM. A positive angle results in forward movement while a negative angle results in backward movement
Right Joystick	Horizontal Joystick Angle	Channel 1	Channel 2 (right motor) Channel 5 (left motor)	The horizontal joystick angle controls the left and right rotational movement of the TCM
Aux Switch D	Up or Down	N/A	Channel 1 Channel 2 (right motor) Channel 5 (left motor)	Aux switch D controls the power provided to the motors. Having the switch down will provide 100% power to the motors and will be referred to as “Sport Mode”. Having the switch up will only provide 20% power to the motors and will be referred to as “Normal Mode”.

Figure 11: RC Controller



The outputs of Sub-System 2 are the resulting motor control actions resulting from the user inputs. Noted below are the resulting motor behaviours, the RC channel control inputs are communicated through, the associated input that caused the output, and the system behavioural description.

Table 15: Functional Description of RC Receiver Outputs

Output	RC Channel	Associated Input	Description
Right Propellor Rotational Direction	Channel 2	Vertical Joystick Angle	The right propellor will rotate clockwise resulting in a forward motion when the joystick has a positive vertical angle. Similarly, the right propellor will rotate counterclockwise resulting in a backward motion when the joystick has a negative vertical angle.
Left Propellor Rotational Direction	Channel 5	Vertical Joystick Angle	The left propellor will rotate clockwise resulting in a forward motion when the joystick has a positive vertical angle.

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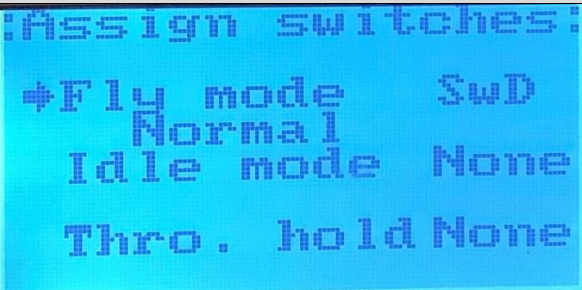
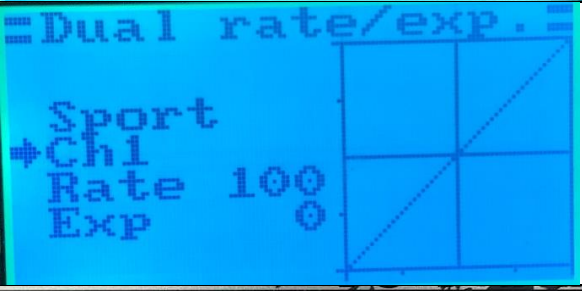
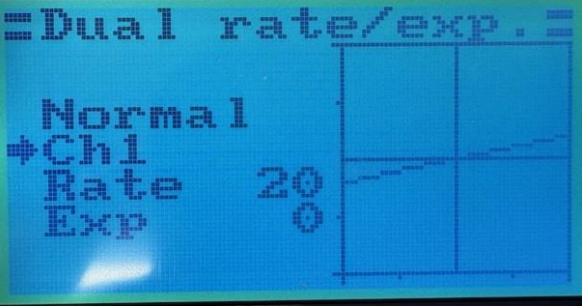
Output	RC Channel	Associated Input	Description
			Similarly, the left propellor will rotate counterclockwise resulting in a backward motion when the joystick has a negative vertical angle.
Isolated Propellor Movement	Channel 2 Channel 5	Horizontal Joystick Angle	When the joystick is moved horizontally to the left, the right motor will move in isolation, resulting in a counterclockwise rotational movement of the TCM (i.e. turning left). Similarly, when the joystick is moved horizontally to the right, the left motor will move in isolation, resulting in a clockwise rotational movement of the TCM (i.e. Turning right).
Motor Speed	Channel 2 Channel 5	Aux Switch D	When the switch is up the motors will operate at 20% power, resulting in a slower boat speed for garbage collection operations. When the switch is down the motors will operate at 100% power, resulting in a faster boat speed for quick boating maneuvers. Exact speeds have yet to be determined.

Design

Note that the Flysky FS-i6X Digital Proportional Radio Control System Instruction Manual (Flysky RC model technology co., ltd, 2016) was referenced to program the RC controller.

Table 16: RC Controller Settings and Explanation

Programming Description	RC Controller
Mix 1 allows motor synchronization. The slave channel (channel 5) is programmed to mimic the master channel's (channel 2's) signals in accordance with the vertical joystick input. An offset of -45% was determined experimentally to properly synchronize the activation of the two motors. As a result, when forward motion is requested, both motors rotate clockwise. Likewise, when backward motion is requested, both motors rotate counterclockwise.	
Mix 2 isolates the movement of the right motor. The master channel (channel 1) corresponds to the horizontal movement of the right RC joystick. Channel 2 (corresponding to the right motor) is programmed to mimic the "positive" signals of channel 1, which are left joystick movements. Therefore, when the joystick is moved to the left, the right motor turns on synchronously. As a result, when the user moves the controller joystick to the left, the TCM turns left.	
Mix 3 isolates the movement of the left motor. The master channel (channel 1) corresponds to the horizontal movement of the right RC joystick. Channel 2 (corresponding to the left motor) is programmed to mimic the "negative" signals of channel 1, which are right joystick movements. Therefore, when the joystick is moved to the right, the left motor turns on synchronously. As a result, when the user moves the controller joystick to the right, the TCM turns right.	

Programming Description	RC Controller
This programming page allows auxiliary switch D to change operation modes of the system.	
Sport mode has been programmed to provide 100% of the normal operating power for the motors.	
Normal mode has been programmed to provide 20% of the normal operating power to the motors.	

Undesired Event Handling

Normal system operations of sub-system 3 are as described in the design details above. In the event of a signal disruption between the RC transmitter and receiver due to the TCM exceeding the RC communication range, power failures, water damage, wire disconnections, or any other unforeseen events, a motor failsafe shut off is triggered, automatically disconnecting power to the motors. Additionally, the RC controller will make a designated noise, inform the user of the signal disconnection and results in a motor shut off. Upon reconnection of the signal, the motors are automatically reactivated and can be controlled immediately without any manual intervention needed.

Sub-System 3: MCU

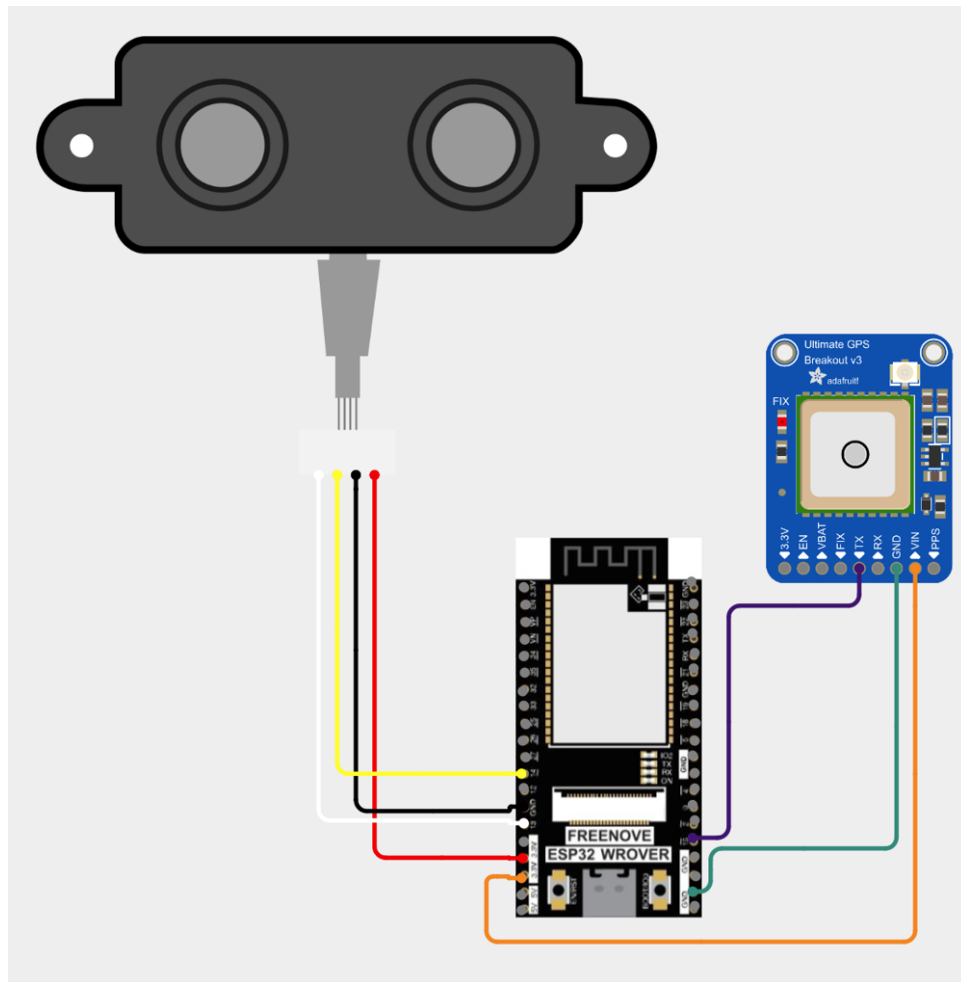
Description of Behaviour

The purpose of this subsystem is to facilitate communication between the three peripheral devices (camera, GPS, and proximity sensor) and the UI. These three peripherals interact with the surrounding environment of the boat, and existing libraries will be used to gather raw data from said peripherals.

Note that data parsing and validation will be handled within the UI subsystem, thus the only code that is included in this section will be the initialization of both peripherals. Also, we use the MicroPython machine library to handle UART communications with the GPS and proximity sensor, and the MicroPython camera driver to handle camera communications. These are the interfaces that Freenove recommends using when handling UART and camera communications (Freenove).

Schematic

Figure 12: Sub-System 3 Electrical Schematic



Note the following:

1. According to the GPS datasheet (Adafruit Ultimate GPS), the common collector voltage (VCC) range is 3.0-5.5V, with a typical value of 3.3V (Beitian Co., Limited, 2015). Thus, it is connected to 3.3V VCC.
2. The GPS uses a serial communication protocol to communicate with the host (MCU) (Beitian Co., Limited, 2015), which includes a transmit (TX) and receive (RX) line. To the best of our knowledge, the serial communication protocol used is UART (Universal Asynchronous Receiver/Transmitter).
3. The following pins can be used for the UART bus: 0, 2, 4, 5, 9, 10, 12-19, 21-23, 25, 26, 34-36, and 39. It is recommended by the manufacturers, based off demo scripts, that pins 13 and 14 or 5 and 2 (RX and TX, respectively) can be used for UART (Freenove). Our group has thus chosen to use pins 13 and 14 for communication with the proximity

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sensor, and pins 15 and 12 for communication with the GPS (as there is no need for the GPS to receive data from the MCU, the RX pin on the MCU and TX pin on the GPS is the only connected serial data pin).

4. The camera is connected to the MCU via a ribbon cable (Freenove).

Inputs & Outputs

The inputs to the MCU subsystem are the local environment around the boat, and the geographic location of the boat. There are no software inputs as the purpose of this subsystem is to take environmental inputs and properly format the data to be sent to the UI.

The outputs of the MCU subsystem are as follows:

Table 17: MCU System Outputs

Output	Source	Format	Purpose
GPS Data	GPS module (MCU)	NMEA 0183	Display system position/coordinates
Proximity Sensor Data	Proximity sensor module (MCU)	Distance data (4 bytes)	Display distance from trash to the boat hull so user knows the boat is collecting trash
Camera Stream	Camera module (MCU)	Video stream	Visual observation of the system's surroundings

Note that this table matches the UI inputs table as all outputs from the MCU subsystem are to be displayed on the UI.

Software Design: Initialization

GPS and Proximity Sensor

Command to initialize a UART object: UART (id, baudrate, bits, parity, rx, tx, stop, timeout)

(Freenove)

Table 18: GPS and Proximity Sensor Initialization Parameters

Parameter	Description	GPS	Proximity Sensor	Justification
id	Serial number	2	1	Can only be 1 or 2. Order is trivial.
baudrate	Baud rate	9600	9600	Matching baud rates ensures synchronized and reliable data transmission across both peripherals.
bits	Bits	8	8	Both devices transmit 1 byte of data at a time.
parity	Odd/even checking	None	None	Following both peripherals' specifications.
rx	MCU receive	15	13	From above schematic.
tx	MCU transmit	12	14	From above schematic. Note that the MCU TX pin for the GPS is not connected.
stop	Number of stop bits	1	1	Following both peripherals' specifications.

Camera

At the moment, camera initialization settings are to be determined. However, our group intends to start with the following command to initialize the camera:

```
camera.init(0, format=camera.JPEG, fb_location=camera.PSRAM)
(Freenove)
```

These settings ensure that the camera output is in JPEG format and enables the use of PSRAM, which will allow the camera to capture higher quality images. However, these parameters are subject to change, and additional parameters may be changed from their default values if our team finds that their default settings are insufficient for the user experience.

Sub-System 4: UI

Inputs & Outputs

Table 19: UI System Inputs

Input	Source	Format	Purpose
Camera Stream	Camera Module (MCU)	Video stream	Visual observation of the system's surroundings.
Object Distance	Proximity Sensor (MCU)	Distance (mm)	A secondary system for trash detection in the case of video stream lag.
GPS Data	GPS Module (MCU)	NMEA 0183	Display system position/coordinates.

Table 20: UI System Outputs

Output	Destination	Purpose
Live Camera Feed	UI	Allows user to identify the trash and provides situational awareness.
Object Distance Indication	UI	Displays proximity sensor measurements to identify nearby floating debris.
Proximity Alert Indication	UI	Warns the user when an object is detected. Shows status of <i>Clear</i> when an object is of a distance greater than 500mm from the sensor. Shows status of <i>Object Detected</i> when an object is less than or equal to 500mm from the sensor.
Latitude / Longitude Coordinates	UI	Provides the user with the latitude and longitude coordinates of the system and updates in real-time.
Number of Satellites Used	UI	Provides the user with the number of satellites being used by the GPS module, so the user can know how accurate the coordinates are.
Map Display	UI	Reporting location of the system on a live map with a moving marker.

Description of Behaviour

Upon startup, the UI initializes map view, proximity data and video display components. When valid GPS data is received from the GPS module, the UI updates the map marker to reflect the system's current position. If GPS data becomes unavailable or invalid, the UI retains the last known position and displays a warning indicator. The live camera feed is displayed continuously.

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The proximity sensor shows the status of *Clear* or *Object Detected* depending on if there is an object flowing into the TCM.

Software Design

The UI software is decomposed into display and data-handling modules to ensure clear separation between data acquisition, validation, and visualization. As the UI is observational only, no control logic is implemented.

Module Guide

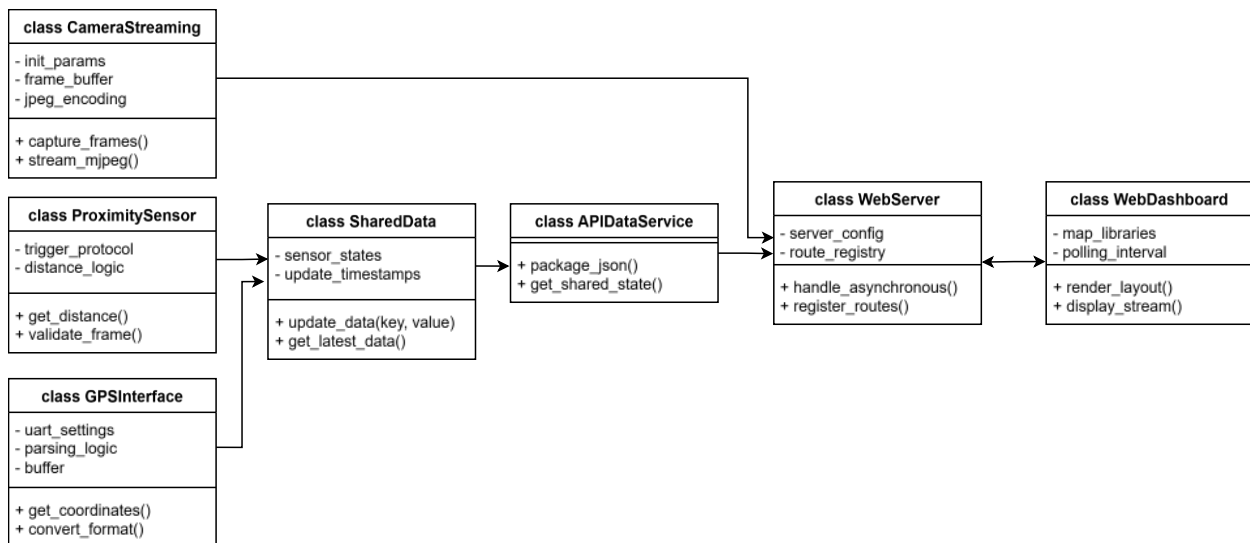
Table 21: UI System Module Guide

Module	Responsibility	Module Secrets (Information Hiding)
Camera Streaming Module (.py)	Captures frames from the ESP32 camera and streams them to connected clients as an MJPEG video feed.	Camera initialization parameters, frame buffering strategy, JPEG encoding, and multipart streaming implementation.
Proximity Sensor Interface Module (.py)	Communicates with the proximity sensor to retrieve distance measurements used to detect nearby floating debris.	Sensor trigger/response protocol, frame validation, checksum verification, and distance calculation logic.
GPS Interface Module (.py)	Receives and parses NMEA messages from the GPS module via UART and extracts location information such as latitude, longitude, and satellite count.	UART communication settings, NMEA sentence parsing logic, coordinate format conversion, and GPS data buffering.
API Data Service (.py)	Packages current sensor data (GPS coordinates, satellite count, and object distance) into JSON responses for the dashboard.	Data formatting, coordinate conversion to decimal degrees, and retrieval of shared sensor state information.
Shared Data Module (.py)	Maintains the latest sensor readings and makes them accessible to both the data acquisition processes and the web server services.	Internal data structures used for storing sensor states and update timestamps.
Web Server Module (.py)	Hosts the embedded HTTP server and registers routes for	Server configuration, request routing, asynchronous handling,

Module	Responsibility	Module Secrets (Information Hiding)
	the dashboard, camera stream, and sensor data API.	and resource management for low-memory environments.
Web Dashboard Module (.html)	Displays the system map, live camera feed, and proximity sensor readings in the user's web browser.	Rendering logic, map visualization libraries, polling intervals, and layout configuration.

Module Decomposition

Figure 13: UML Class Diagram



Module Internal Design (MID)

Note: The names of the modules in the Module Guide differ from the modules listed in the GitHub. This is to provide an abstract view of the program to better convey the functionalities of the various modules through the Module Guide. Please refer to the *Rev 1* folder in the RIPPLE repository.

GitHub Repository: <https://github.com/Thuvarakaa/RIPPLE.git>

Pseudocode

CLASS SharedData:

```
// Attributes for thread-safe state management
sensor_states = {"distance": 0, "lat": 0, "lon": 0, "sats": 0}
update_timestamps = 0
```

```
FUNCTION update_data(key, value):
    Set sensor_states[key] = value
    Update update_timestamps to current time
```

```
FUNCTION get_latest_data():
    RETURN sensor_states
```

CLASS ProximitySensor:

```
// Logic for UART-based distance sensing (13/14)
```

```
FUNCTION get_distance():
    FLUSH UART buffer
    SEND trigger_protocol byte (0x55)
    IF 4 bytes received within timeout:
        RETURN validate_frame()
```

```
FUNCTION validate_frame():
    // Checksum: (Header + H + L) & 0xFF
    IF header == 0xFF AND checksum_is_valid:
        distance_logic = (HighByte << 8) + LowByte
        RETURN distance_logic
```

CLASS GPSInterface:

```
// Logic for NMEA parsing (Pin 15)
FUNCTION get_coordinates():
    WHILE data available on UART:
        READ byte and update MicropyGPS parser
```

```
FUNCTION convert_format():
    // Convert Degrees/Minutes to Decimal Degrees
    lat_dd = degrees + (minutes / 60)
    IF hemisphere == 'S': lat_dd = -lat_dd
    lon_dd = degrees + (minutes / 60)
    IF hemisphere == 'W': lon_dd = -lon_dd
    RETURN lat_dd, lon_dd
```

CLASS CameraStreaming:

```
// Logic for ESP32-S3 Camera and MJPEG stream
FUNCTION capture_frames():
    INITIALIZE camera with init_params (pins, freq, PSRAM)
```

```
WHILE active:
    buffer = capture image from sensor
    YIELD jpeg_encoding of buffer
```

```
FUNCTION stream_mjpeg():
    START multipart/x-mixed-replace HTTP response
    FOR EACH frame FROM capture_frames():
        WRITE frame to response stream
```

CLASS APIDataService:

```
// Logic for providing JSON telemetry to the dashboard
FUNCTION get_shared_state():
    raw_data = SharedData.get_latest_data()
    // Process raw GPS through convert_format()
    RETURN processed_telemetry

FUNCTION package_json():
    data = get_shared_state()
    RETURN JSON_STRING(data)
```

CLASS WebServer:

```
// Orchestration of the Picoweb app and background tasks
FUNCTION register_routes():
    ROUTE "/" TO WebDashboard.render_layout()
    ROUTE "/api/data" TO APIDataService.package_json()
    ROUTE "/video" TO CameraStreaming.stream_mjpeg()

FUNCTION handle_asynchronous():
    INITIALIZE WiFi (SSID: "Vedashri", PASSWORD: "vedashri18")
    START background_thread for ProximitySensor and GPSInterface
    START WebServer.run()
```

CLASS WebDashboard:

```
// Frontend logic running in the browser
FUNCTION render_layout():
    SERVE index.html from flash memory
    LOAD map_libraries (Leaflet.js)

FUNCTION display_stream():
    SET polling_interval = 1000ms
    EVERY interval:
        FETCH /api/data
        UPDATE map marker and text stats
        IF distance < 500: TRIGGER alert_active
```

Undesired Event Handling

Table 22: Undesired Event Handling

ID	Event	Response	Rationale
E1	UI loses connection to the GPS, camera modules, and the proximity sensor	Warning on the UI to inform users of issues with data retrieval. The UI is able to identify this issue via the timeout feature.	This makes the user aware that there is an issue in the Wi-Fi connection, and they can attempt to reconnect with the modules.
E2	Net rips/detaches from Boat Hull	User retrieves the TCM and manually switches the net out and recollect the “leaked waste”	Switching out the net and recollecting the waste ensures the waste collection process is completed and prevents the system from producing excess waste.
E3	Out of RC range	RC will alert the user that the TCM is moving out of range, and the user should bring TCM closer to the RC	User won’t have to constantly keep track of the distance of the TCM, they will only need to prevent it from going out of range.
E4	Loss motor control	Emergency stop: motors will stop moving once the connection is loss.	This will allow the TCM to stay in one place while the user manually retrieves the TCM and check for damages.

Timing Constraints

The following requirements define the performance expectations of our surface-water trash collector, specifying how quickly the system must detect, collect, and dispose of floating debris in West Pond.

Table 23: Performance Timing Requirements

RID	Description	Likelihood of Changing
R8	Detection Response Time: Using camera visualization, the user should be able to detect a standard floating object approaching the TCM within 5 seconds.	Low
R9	Collection Cycle Time Per Item: The TCM should complete a full collection time under normal conditions (user detect, align, capture, store) for a properly sized item in less than 20 seconds.	Medium
R10	Battery Efficiency Cycle: The system shall operate continuously for at least 1 hours before it needs to be recharged.	Low
R11	Alignment Time: The TCM should properly align the net to intercept an object within 10 sec from 2 meters away from the net.	Medium
R12	Direction Change Time: The TCM should be able to change direction by 90° angle in 3 seconds.	Low
R13	Thrust to Drag Response Time: The propulsion mechanism shall generate sufficient thrust to overcome drag and provide forward motion within 3 seconds of motor activation from rest. <i>Rationale:</i> This ensures that the system can overcome water and collected trash resistance quickly, preventing stall at low speeds.	Low
R14	Motor Response Time: Each of the motors should reach 100% of the user input speed within 3 seconds of receiving the control signal.	Low
R15	GPS Response Time: The delay between receiving GPS data from the GPS and the ESP32 main control loop processing shall be ≤ 1 second	Low

System Validation

Table 24: Safety & Security Requirements

RID	HID	Description	Contribution To Problem Solution
SSR1	H7	Waterproof system: The electrical components in the TCM shall have waterproof housings to prevent damage from splashes.	The goal of the system is to collect trash on top of water, so having waterproof components ensures that the intended use environment won't damage the product.
SSR2	H7 H10 H11 H12	Emergency stop: If TCM loses connection with RC for more than 5 seconds, power supply shall be cut off.	When RC communication is lost, the motors will otherwise continue to run at the last communicated speed. If an error occurs, an emergency shut off will ensure that the environment is minimally damaged in the event of this disruption.
SSR3	H14	Alert for Environmental interference: TCM should alert user when motor is moving but GPS coordinates are not changing. <i>Rationale: Allows user to check for environmental interference (animals / branches / rocks / stuck / etc.)</i>	Having this alert is important for reducing the chances that unseeable obstacles (such as ones underwater) will cause the boat to become stuck for a prolonged period without the remote user being aware.
SSR4	H3 H4	Communication Range: TCM should alert user through UI when TCM is within 3 meters away from out-of-range boundaries.	Alerts the user to move the TCM closer to avoid a loss of communication, which ensures it can still be operated on the surface of water without the need for manual retrieval.
SSR5		User Manual: User manual shall be created to prevent operational hazards.	This will ensure the operator is properly informed on how to safely operate the system and notify them of potential hazards, enhancing safe and effective operation of the system.
SSR5.1	H20	<i>User must operate system in ideal weather and water conditions.</i>	<i>The TCM will be damaged if used in high wind or stormy environments. The user must</i>

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RID	HID	Description	Contribution To Problem Solution
			<i>be aware when not to use the system to reduce the chances of damage.</i>
SSR5.2	H9	<i>User must ensure that camera is clean/ view is not obstructed before operation.</i>	<i>The user must have a clear camera to locate trash that needs to be collected or obstacles to avoid.</i>
SSR5.3	H17	<i>User must monitor the camera footage rigorously during operation.</i>	<i>The user must be able to locate trash or obstacles in a timely manner to maneuver the TCM accordingly (to either align the boat hull to collect or avoid obstacles).</i>
SSR5.4	H16 H18	<i>User must check system upon recollection for physical damage.</i>	<i>The user must check for issues in the system to ensure that the boat hull can be used again for subsequent collections, as pollution is an ongoing issue not solved by a single collection.</i>
SSR5.5	H19	<i>User must be aware of the amount of trash collected by the system to mitigate overflow.</i>	<i>The user must be aware of the amount of trash collected so that the boat does not exceed operational limits and becomes pollution itself when it sinks due to exceeded weight limits and/or trash escapes collection net due to overflow.</i>
SSR5.6	H13	<i>User must be aware of max turning radius.</i>	<i>The user must have a good understanding of the turn radius to ensure that the TCM doesn't get stuck in a position that it can't move out of.</i>
SSR6	H22	Speed limit: Maximum operating speed of TCM shall not exceed 5 m/s.	The speed is limited to 5m/s to ensure that the TCM remains controllable and gives the user an appropriate amount of time to react to obstacles.

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RID	HID	Description	Contribution To Problem Solution
SSR7	H21	Sound insulation: System should be sound insulated as to not disturb the natural environment.	The system should not interfere with wildlife nearby and other stakeholders such as residents.
SSR8	H1 H2 H5 H6 H8 H10 H15	Component validation: All electrical components of system shall be tested prior to system assembly to validate specifications.	To ensure that the system is fully safe and operational before use.

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Table 25: Behavioral Requirements

RID	Description	Contribution To Problem Solution
R1	Floatation: The boat hull shall be floating during the entirety of the operation cycle.	Incorporates a buoyancy-driven flotation system to prevent sinking during all operational conditions.
R2	Trash Collection: The TCM shall securely collect trash floating on the surface of small bodies of water.	The TCM consists of a net that will collect the trash when in forward motion.
R3	Optimal Operation: The TCM shall operate reliably on the surface of water carrying all associated components and any collected trash up to 1000 cm ³ .	Ensures surface water trash is collected and that the net provides sufficient capacity for up to 1000 cm ³ of debris.
R4	System Movement: The TCM's movement should be remotely controlled by the user.	Provides ease of use and user control during operation.
R5	Turning Radius: The system should allow forward movement and changes of direction with a 30 cm turning radius.	Allows the system to move around efficiently confined areas and navigate around obstacles.
R6	Location Detection: The TCM component should be remotely located by the user using GPS coordinates.	Allows the user to track and locate the TCM in real time if the system is out of eyesight.
R7	Visualization: The system shall provide the user with live video footage of the front view of the TCM.	Provides user with real time visual feedback to assist the user in navigation and monitoring.

Table 26: Performance Timing Requirements

RID	Description	Contribution to Problem Solution
R8	Detection Response Time: The camera should be able to detect a standard floating object approaching the TCM within 5 seconds.	Timely detection is critical for surface-level trash collection because the system relies on visual feedback to initiate alignment and capture.
R9	Collection Cycle Time Per Item: The TCM should complete a full collection time under normal conditions (camera detect, align, capture, and store) for a properly sized item in less than 20 seconds.	Ensures operational efficiency and allows the system to collect multiple items within a deployment window.
R10	Battery Efficiency Cycle: The system shall operate continuously for at least 1 hour before it needs to be recharged.	A minimum one-hour operational duration ensures practical field usability by limiting the frequency of battery swaps or recharges.
R11	Alignment Time: The TCM should properly align the net to intercept an object within 10 sec from 2 meters away from the net.	Time efficient alignment is necessary to compensate for object drift caused by water currents and vehicle inertia.
R12	Direction Change Time: The TCM should be able to change direction by 90-degree angle in 3 seconds.	This is useful in avoiding debris and obstacles.
R13	Thrust to Drag Response Time: The propulsion mechanism shall generate sufficient thrust to overcome drag and provide forward motion within 3 seconds of motor activation from rest. Rationale: This ensures that the system can overcome water and collected-trash resistance quickly, preventing stall at low speeds.	Ensures that the system can reliably transition from rest to motion despite hydrodynamic drag and resistance from the net that has accumulated trash.
R14	Motor Response Time: Each of the motors should reach 100% of the user input speed within 3 seconds of receiving the control signal.	Affects controllability of the system and system responsiveness. Defining a response threshold ensures that user commands translate into a timely mechanical response.

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RID	Description	Contribution to Problem Solution
R15	GPS Response Time: The delay between receiving GPS data from the GPS and the ESP32 main control loop processing shall be ≤ 1 second	Limiting GPS data latency ensures that position of information remains relevant for navigation.

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Table 27: Hardware Requirements

RID	Description	Contribution to Problem Solution
R16	Operation Temperature: The system should operate reliably between 3 degrees and 28 degrees Celsius.	This will ensure the system can be used all year round (excluding winter) with varying air and water temperatures.
R17	UV-Stabilization: The TCM's plastic enclosures, plastic structure, and net must be UV-stabilized to limit degradation.	Having UV stabilized components will increase the lifespan of the system as West Pond is outdoors and any trash collection on its waters will occur under daylight.
R18	Operation Levels: The system's safe operation is guaranteed up to wind $\leq$ TBD, wave height $\leq$ TBD, current $\leq$ TBD m/s.	This will ensure that the system can withstand a variety of environmental conditions within a natural pond environment.
R19	Dynamic Load Operation: Net frame and attachments shall withstand the impact of dynamic loads (drag force) and hold up to 500g continuous load in the TCM.	This will ensure the system does not get stuck as a result of collecting too much trash, effectively confirming that the system itself will not contribute to surface water waste by getting stuck.
R20	Maintenance Accessibility: The system's compartments (battery, electronics, motors, GPS tracker, camera) shall be accessible for inspection/maintenance.	Serviceability is an important part of the system's design as operators should be able to maintain the system for a long lifespan, furthermore, ensuring the system does not contribute more waste than it collects.
R21	Component Mounting: GPS Tracker and camera shall be mounted on a raised, unobstructed mount.	This refers to RF and visual obstruction and ensures both peripherals can operate without interference so they can report accurate data to the operator.
R22	Camera Specifications: Camera viewport shall be optically clear. Camera must support required resolution for detection.	If the viewport is obstructed or the camera's resolution is too low, the operator cannot use the system from a distance, one of its main features.
R23	Trash Loss Prevention: The TCM should be in continuous forward motion to prevent the loss of trash collected.	As the purpose of the system is to collect surface trash, this helps the system most effectively accomplish its purpose by not

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RID	Description	Contribution to Problem Solution
		letting any collected trash fall out of its net.
R24	Connection Status: The system shall have sufficient wireless connection between the ESP32 and the controller.	This is critical to enable the remote operation of the system. If any lag is present, it could negatively impact the system's ability to collect trash.

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Table 28: Software Requirements

RID	Description	Contribution to Problem Solution
R25	Real-time Control: The remote controller shall enable real-time control of the system's movement, allowing independent operation of the two motors.	This allows the operator to precisely maneuver the system toward specific pieces of floating trash, ensuring no debris is missed due to lag or unresponsive steering.
R26	Sensor Integration: The software shall interface with the onboard GPS module and live camera to collect position and visual data.	By combining GPS and visual data, the system can provide the visual confirmation needed to distinguish between trash and aquatic life.
R27	Data Communication: The system shall transmit data from the microcontroller to the external UI.	This bridges the gap between the system and the operator, ensuring that updates from the sensors are known in real-time.
R28	UI Support: The UI should display real-time coordinates and live video feed.	The UI transforms raw data into a usable "cockpit" view, enabling the operator to navigate safely in murky waters or across long distances where the system isn't visible.
R29	Synchronization and Timing: The software shall ensure synchronized operation between remote controller data, motor control, and data transmission.	Ensuring motor commands and sensor data happen in sync prevents delayed responses, which is vital for avoiding obstacles in the water.
R30	Modularity: The software shall be structured in modules (motor control, GPS, camera, UI) to simplify development, testing, and maintenance.	By isolating the code, the team can troubleshoot sub-system failures without accidentally breaking something else, leading to more reliable maintenance.
R31	Extensibility: The software environment should allow integration of additional modules or functionalities such as mapping, analytics, or remote monitoring.	This allows the system to eventually carry more sensors or added functionality, such as object detection logic to clean larger areas more efficiently.

System Verification

Testing Limitations

The testing of the surface trash collecting mechanism was conducted using a circular pool with a diameter of 3m. Due to the limited testing apparatus available, the requirements that are out of scope to test currently in these environmental conditions are listed in the table below.

Table 29: Requirements That Will Not Be Tested

RID	Description	Reasoning
SSR3	Alert for Environmental Interference	This requirement is out of scope as we do not intend to implement a sensing mechanism on the motors to detect whether they are moving.
SSR4	Communication Range	This requirement is out of scope as the pond we intend to test at is smaller than the MCU's communication limit.
SSR5	User Manual	You cannot test a user manual. It is simply there to inform the user of safe operation conditions and probable hazards.
R16	Operation Temperature	The full integrated system cannot be tested in different temperature environments, as we do not have access to a thermostat in the space where we are testing.
R17	UV-Stabilization	Testing whether the mechanical components of the system are UV-stabilized is out of our scope.
R18	Operation Levels	We cannot test the system outside due to the cold weather and do not have access to facilities which simulate wind or currents.
R23	Trash Loss Prevention	As our pool for testing is not large enough for the system to maintain continuous forward motion, this requirement is out of scope.
R30	Modularity	It is not possible to test whether the UI is modular, as this is a requirement for UI <i>design</i> rather than operation.
R31	Extensibility	It is not possible to test whether the UI allows for the integration of additional modules and functionalities, as this is a requirement for UI <i>design</i> rather than operation.

Integrated System

Objective

The objective of this verification process is to evaluate the performance of the integrated system to determine if current design solutions fulfill the requirements defined in *Validation*. Specifically, these tests will confirm that the design solution can allow the user to locate floating trash, collect it, and continue to maneuver flexibly in the water while keeping collected trash secure in the collection net in a timely manner, and at appropriate speeds.

Data & Test Plan

Table 30: Integrated System Test Plan & Results

Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
1	SSR6	Max input power to motors	Speed of <5m/s	Speed of ~1.5m/s	Pass
2	R1	Boat left in pool with all components for an hour	Boat still floating after elapsed time	Boat was still floating after elapsed time	Pass
3	R2	1 floating piece of trash in pool	1 piece of trash in TCMs collection net	1 piece of trash in TCMs collection net	Pass
4	R2	Boat left floating with 1 piece of trash for 5 minutes	1 piece of trash remains in TCMs collection net	1 piece of trash remains in TCMs collection net	Pass
5	R3/R4/ R19/ R23/ R25	Forward joystick input with 1000 cm ³ of trash in collection net	Forward movement with no lost trash	Forward movement with no lost trash	Pass
6	R3/R4/ R25	Left joystick movement with 1000 cm ³ of trash in collection net	Left turn with no lost trash	Boat turned left with no lost trash	Pass
7	R3/R4/ R25	Right joystick movement with 1000 cm ³ of trash in collection net	Right turn with no lost trash	Boat turned right with no lost trash	Pass

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Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
8	R4/R25	Backwards joystick movement	Boat reverses	Boat reverses	Pass
9	R5	Held right joystick movement	Right turn with radius <30cm	Right turn with radius of ~0 (boat turned on a dime)	Pass
10	R9/R11	A piece of trash is left in the testing pool for a user in another room to collect	Collection occurs in <20s, detection and alignment occurs in <10s	Collection occurred in 18 seconds with alignment occurring in 8 seconds	Pass
11	R12	Held left joystick movement	90-degree direction change in <3s	90-degree direction change in ~2s	Pass
12	R13/R14	Forward joystick movement while boat is stationary	Forward boat movement in <3s	Forward boat movement in <1s	Pass
15	R20	User tries to reach all components of fully assembled system	All components can be reached	All components can be reached	Pass

Results

The boat was able to collect trash securely, maneuver flexibly with a large, collected load while meeting all defined special and time constraints. There were no major concerns with the behaviour of the system throughout the integrated testing.

Sub-System 1: Boat Hull

Objective

The objective of this validation and verification process is to evaluate whether the boat hull successfully fulfills all required functional and operational tasks in the pond environment. This includes confirming reliable propulsion, stability, and maneuverability within a body of water. The testing aims to verify that the system meets design requirements, performs consistently, and operates safely. Results from these tests will be used to identify any performance gaps and inform further design improvements.

Data & Test Plan

Table 31: Boat Hull Subsystem Test Plan & Results

Test ID	RID	Inputs	Expected Values	Output	Pass/Fail
1	R1	Boat assembled with all the components Place boat in pool, leave for 1 hour	Boat remains afloat after 1 hour No major water ingress	The boat floated for 1hr	Pass
2	R2	Place 4 floating pieces of trash in pool Operate TCM Allow boat to run until collection attempt completes	All 4 pieces of trash already collected Trash is contained	Collected 4 pieces of trash and stayed in the net	Pass

Results

All test cases were successfully completed and met their expected outcomes. The boat remained floating with all components inside for the required duration, confirming sufficient buoyancy and stability. Additionally, the TCM effectively collected all required trash and retained it during operation without loss. Since all tests resulted in a pass, the system is considered to meet the specified requirements and perform reliably under the tested conditions for the mechanical structure.

Sub-System 2: Motor Control

Objective

The objective of this validation process is to evaluate the performance of the motor specific tasks to determine if current design solutions fulfill the requirements defined in *Validation*.

Specifically, these tests will confirm that the motors will shut off in the event of an unforeseen RC communication disruption, all motor components perform as expected before being integrated with the boat, and the system is able to operate for over 1 hour.

Data & Test Plan

Table 32: Motor Subsystem Test Plan & Results

Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
1	SSR2	Receiver and Transmitter lose connection for >5s	Motors shut off	Motors Shut off	Pass
2	SSR8	Forward joystick movement	Motors rotate clockwise synchronously	Motors rotated clockwise synchronously	Pass
3	SSR8	Backward joystick movement	Motors rotate counterclockwise synchronously	Motors rotated counterclockwise synchronously	Pass
4	SSR8	Left joystick movement	Right motor rotates in isolation	Right motor rotated in isolation	Pass
5	SSR8	Right joystick movement	Left motor rotates in isolation	Left motor rotated in isolation	Pass
6	R10	Continuous motor movements for 1 hour	Motor operates for >1 hour	After installing motor cooling jackets, the motors have continuous movements for more than 1 hour.	Pass

Results

After testing it was confirmed that the motor subsystem performed almost as expected. The testing revealed that with prolonged use the motors heat up excessively and will call for additional cooling elements such as a cooling motor jacket for longer use cases.

Sub-System 3: MCU

Objective

The objective of this validation and verification process is to evaluate the performance of GPS coordinate acquisition and display to determine if current design solutions fulfill the requirements defined in *Appendix A*. Specifically, these tests will determine if the GPS subsystem can accurately locate the TCM and display this information to the user with minimal time delay.

Data & Test Plan

Table 33: MCU Subsystem Test Plan & Results

Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
1	R26/ SSR8	GPS connected to the MCU as per schematic	MCU can receive data from the GPS within <1s	It did not work initially. However, once we swapped the RX/TX lines (as per UART protocol) we were able to receive data in <1s.	Pass
2	R6/ R15/ R26	GPS left at known coordinates while connected to MCU and covered by PLA enclosure	GPS can find a fix in <15s, maintains its fix, and position is as accurate as the iPhone compass app	GPS can find a fix in <15s, maintains its fix, and position is more accurate than the iPhone compass app	Pass

Results

During testing, we realized that we had a hard time finding and maintaining a fix with the BN-220 GPS, and as we were testing inside Thode and GSB, we realized the signal of our GPS was obstructed. We ended up purchasing another GPS, the Adafruit Ultimate GPS Breakout with GLONASS + GPS (which has a PA1616D antenna). This module allowed us to add backup power storage for the GPS clock, ensuring it can maintain location data while the MCU is disconnected from power. As a result, we were able to find and maintain a fix easily with the new GPS board when testing unobstructed on BSB field, even when placing the module inside of its PLA enclosure (as PLA is RF-transparent).

Sub-System 4: UI

Objective

The objective of this verification testing process is to evaluate the performance of the UI in terms of displaying the received camera feed in the form of a livestream and GPS data in the form of longitudinal/latitudinal coordinates as well as a visual marker on a map. The communication and data reception is facilitated over Wi-Fi.

Data & Test Plan

Table 34: UI Subsystem Test Plan & Results

Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
1	R25/ R26/ R27/ R28	UI connects with hotspot	Terminal on Thonny displays the same IP address as the browser IP address which displays the UI.	Successfully displays identical IP from Thonny IDE on the browser where the UI is displayed.	Pass
2	R27	UI is displayed on various devices (that are connected to the MCU's IP address)	The UI is accessible on a laptop and phone and tablet device simultaneously.	UI is accessible on the laptop, phone and tablet device simultaneously and show the same data.	Pass
3	R7/ R22/ R24/ R26/ R27/ R28/ R29	Camera communication from MCU to UI	Camera feed is visible on UI.	Camera feed is displayed on UI and frames are consistently updated.	Pass
4	R27/ R28/ R29	GPS coordinates communication from MCU to UI	GPS longitude and latitude are reported on UI and is updated as the system relocates.	The UI displays coordinates for longitude and latitude as well as the number	Pass

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Test ID	RID	Input(s)	Expected Value(s)	Output(s)	Pass/Fail
				of satellites used.	
5	R27/ R28/ R29	Map marker moving on UI	Map marker updates live as the system is maneuvered	The map marker is shown on the map corresponding to the coordinates and moves in real-time.	Pass

Results

The UI satisfies all connection related, GPS, and camera related requirements. In addition, it makes satisfactory connections with and fulfills all objectives for the proximity sensor.